

Statistical modeling

Module 03

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Categorical variables

So far, we only have seen examples where the independent variables x are numerical.

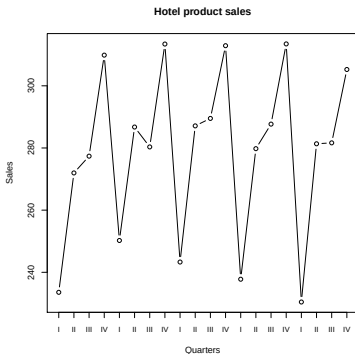
What if one or more of the x_j 's is categorical?

- ▶ Is it still a linear model?
- ▶ How can we write/fit the model?
- ▶ How can we make the prediction?

In fact, the model can be written in the same way. The procedure is first to create so called "dummy variables" and to use them as numerical variables.

Example

Consider the quarterly sales of a hotel product shown below. We ignore trend for simplicity.



The independent variable x is categorical with 4 levels: Quarter (I-IV).

Dummy variables

With 4 levels, we associate 3 dummy variables (nb of levels minus 1). They are defined as indicator whether the observation is in the corresponding quarter:

$$\delta_{II}(x) = \begin{cases} 1, & \text{if } x = II \\ 0, & \text{otherwise.} \end{cases}$$

$$\delta_{III}(x) = \begin{cases} 1, & \text{if } x = III \\ 0, & \text{otherwise.} \end{cases}$$

$$\delta_{IV}(x) = \begin{cases} 1, & \text{if } x = IV \\ 0, & \text{otherwise.} \end{cases}$$

Linear model with dummy variables

The model of the sales y at quarter x is

$$y = \beta_0 + \beta_{II}\delta_{II}(x) + \beta_{III}\delta_{III}(x) + \beta_{IV}\delta_{IV}(x) + e$$

Therefore, the prediction of the price in each quarter are

- ▶ Quarter I: $y(x) = \beta_0$
- ▶ Quarter II: $y(x) = \beta_0 + \beta_{II}$
- ▶ Quarter III: $y(x) = \beta_0 + \beta_{III}$
- ▶ Quarter IV: $y(x) = \beta_0 + \beta_{IV}$

Here Quarter I is the reference level. It is purely arbitrary and any level could be chosen as the reference.

Contrasts

The interpretation of the coefficients are in terms of difference to the reference level. These differences are called contrasts:

- ▶ β_0 : the average sales in Quarter I.
- ▶ β_{II} : contrast between Quarter II and Quarter I.
- ▶ β_{III} : contrast between Quarter III and Quarter I.
- ▶ β_{IV} : contrast between Quarter IV and Quarter I.

How would you estimate the contrast between Quarter II and Quarter III?

Example

R (and any statistical computer program) builds automatically the dummy variables and provides the estimation as usual.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	239.070	2.637	90.67	< 2e-16 ***
QuarterII	42.320	3.729	11.35	4.59e-09 ***
QuarterIII	44.240	3.729	11.87	2.42e-09 ***
QuarterIV	71.920	3.729	19.29	1.67e-12 ***

Residual standard error: 5.896 on 16 degrees of freedom

Multiple R-squared: 0.9596, Adjusted R-squared: 0.952

F-statistic: 126.7 on 3 and 16 DF, p-value: 2.328e-11

In Excel, you need to build the dummy variables by hand.

Dummy variables: general case

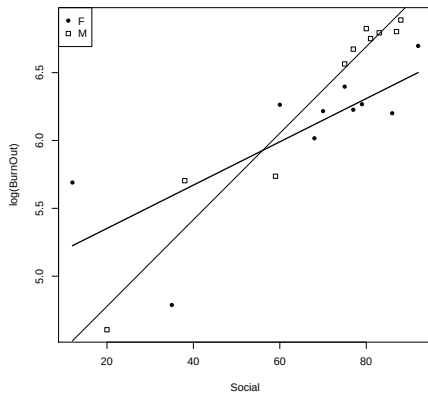
A categorical variable x with m levels $L_1, \dots, L_m$ is coded by $m - 1$ dummy variables

$$\delta_{L_k}(x_2) = \begin{cases} 1, & \text{if } x_2 = L_k, \\ 0, & \text{if } x_2 \neq L_k, \end{cases} \quad k = 2, \dots, m.$$

The level L_1 is the reference level. It has no dummy because the coefficients of the dummies are the contrasts with this level.

Mixing categorical and numerical variables

Example: In a study of the log-burn-out index (y) as a function of the sex (M/F) and a socialization index (x_1). Sex is a categorical variable. Consider two linear models:



Females

$$y = 5.033 + 0.016x_1 + e$$

Males

$$y = 4.14 + 0.031x_1 + e$$

Can we make one regression including the 2 factors Sex and Socialization Index?

Mixing categorical and numerical variables

Let x_2 be the sex with value M or F. The associated dummy variable is defined as

$$\delta_M(x_2) = \begin{cases} 0, & \text{if } x_2 = F \\ 1, & \text{if } x_2 = M. \end{cases}$$

The two models can be written with one single equation:

$$y = 5.033 - 0.893 \times \delta_M(x_2) + \{0.016 + 0.015 \times \delta_M(x_2)\}x_1 + e$$

Dummy variables

Indeed, if $x_2 = M$ then $\delta_M(x_2) = 1$ and

$$\begin{aligned}y &= 5.033 - 0.893 + \{0.016 + 0.015\}x_1 + e \\ &= 4.14 + 0.31x_1 + e\end{aligned}$$

And if $x_2 = M$ then $\delta_M(x_2) = 1$ and

$$y = 5.033 + 0.016x_1 + e$$

Example In R, the output is

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.033075	0.273818	18.381	3.5e-12	***
Social	0.015950	0.003946	4.042	0.000945	***
SexM	-0.892550	0.409303	-2.181	0.044482	*
Social:SexM	0.014934	0.005775	2.759	0.013976	*

Interpretations

The average difference between a female and a male with the same socialization index value, say x_1 , is

$$y_M - y_F = -0.893 + 0.015x_1$$

Thus, the two coefficients associated with the dummy $\delta_M(x_2)$ are the differences between F and M in

- ▶ The contrast in the intercepts
- ▶ The contrast in the slopes

Note that here Female is the reference level: the contrasts are associated with the level Male, the difference is "Male - Female". The reference is arbitrary can be modified by the user.

Matrix notations

Let

$$y = Xb + e$$

where the row X_i is

$$X_i = \left[1 \mid 0 \mid x_{i1} \mid 0 \right], \text{ if } i \text{ is a Female } (x_{i2} = F),$$

and

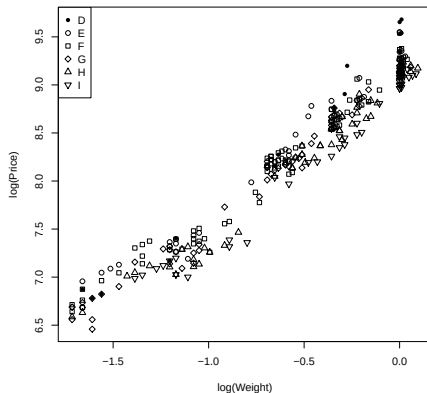
$$X_i = \left[1 \mid 1 \mid x_{i1} \mid x_{i1} \right] \text{ if } i \text{ is a Male } (x_{i2} = M),$$

Overall,

$$X_i = \left[1 \mid \delta_M(x_{i2}) \mid x_{i1} \mid x_{i1}\delta_M(x_{i2}) \right]$$

More than 2 levels

Example: Pricing diamond stones. The response is log-Price (y) and the independent variables are the log-weight (x_1), and the color (x_2). Color is categorical with 6 levels ($D - I$).



More than 2 levels

The analysis in R gives the coefficients:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	9.46974	0.04443	213.157	< 2e-16	***
log(Weight)	1.71716	0.05167	33.234	< 2e-16	***
ColorE	-0.21506	0.05334	-4.032	7.04e-05	***
ColorF	-0.29782	0.04977	-5.984	6.27e-09	***
ColorG	-0.34163	0.05083	-6.721	9.29e-11	***
ColorH	-0.41321	0.05014	-8.241	5.58e-15	***
ColorI	-0.50868	0.05156	-9.867	< 2e-16	***
log(Weight):ColorE	-0.17500	0.06478	-2.702	0.00730	**
log(Weight):ColorF	-0.19427	0.05860	-3.315	0.00103	**
log(Weight):ColorG	-0.17525	0.05884	-2.978	0.00314	**
log(Weight):ColorH	-0.12726	0.06189	-2.056	0.04062	*
log(Weight):ColorI	-0.13292	0.06589	-2.017	0.04456	*

Note: Color D is the reference level.

More than 2 levels

There are 5 dummy variables: no dummy for D (the reference).

$$\delta_E(x_2) = \begin{cases} 1, & \text{if } x_2 = E, \\ 0, & \text{if } x_2 \neq E, \end{cases}$$

$$\delta_F(x_2) = \begin{cases} 1, & \text{if } x_2 = F, \\ 0, & \text{if } x_2 \neq F, \end{cases}$$

$$\delta_G(x_2) = \begin{cases} 1, & \text{if } x_2 = G, \\ 0, & \text{if } x_2 \neq G, \end{cases}$$

$$\delta_H(x_2) = \begin{cases} 1, & \text{if } x_2 = H, \\ 0, & \text{if } x_2 \neq H, \end{cases}$$

$$\delta_I(x_2) = \begin{cases} 1, & \text{if } x_2 = I, \\ 0, & \text{if } x_2 \neq I. \end{cases}$$

There are $(\# \text{levels} - 1)$ dummy variables for a categorical factor.

Interactions between numeric and categorical variables

In the diamond example,

- ▶ Values associated to colors are called contrast. There is no contrast for the reference level.
- ▶ Values combining two factor levels are called interactions. They represent the variation of the one factor effect when the other factor level changes.

The interaction between a categorical factor and a numeric factor is the product between the dummy variables and x .

Interactions between numerical and categorical variables

Example: with diamond data

- ▶ Intercept (9.47) is the intercept for color D.
- ▶ Color E (-0.215) is the contrast of the intercept for color E to D: intercept for color E is $9.47 - 0.215$.
- ▶ $\log(\text{Weight})$ (1.72) is the slope for color D.
- ▶ $\log(\text{Weight}) : \text{ColorE}$ (-0.175) is the slope contrast for color E to D: the slope for color E is $1.72 - 0.175$. It is one interaction term.

Interactions between numerical and categorical variables

Example: with burn-out case

$$\begin{aligned} y = & 291.7 - 142.1\delta_M(x_2) - 4.61x_1 + 0.21x_1\delta_M(x_2) \\ & + 0.10x_1^2 + 0.06x_1^2\delta_M(x_2) + e, \end{aligned}$$

- ▶ -4.61 is the interaction between the sex and the linear term
- ▶ 0.06 is the interaction between the sex and the quadratic term.

Note that -142.1 could formally be seen as the interaction between the sex and the constant term¹.

¹nobody does that...

Interactions between two numerical variables

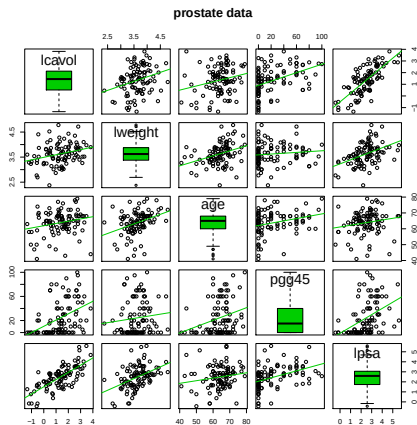
The interaction between a categorical factor and a numeric factor is the product between the dummy variables and x .

In the same way, the interaction between two numerical variables is also the product of the two variables.

Interactions between two numerical variables

Example:

The log-prostate specific antigen (lpsa) is inspected as a function of several variables (log-cancer volume, log-prostate weight, age, etc.).



Interactions between two numerical variables

Model with all predictors and an interaction between pgg45 and lpsa.

```
> mod <- lm(lcavol~lweight+age+pgg45*lpsa, data=prostate)
> summary(mod)
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.849617	0.864081	-0.983	0.328
lweight	-0.190851	0.224882	-0.849	0.398
age	0.014820	0.012172	1.218	0.227
pgg45	0.008882	0.009054	0.981	0.329
lpsa	0.739103	0.105066	7.035	3.62e-10 ***
pgg45:lpsa	-0.001386	0.003011	-0.460	0.646

```
---
[...]
```

Model equation

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_{34}x_3x_4 + e.$$

Estimated to

$$y = -0.850 - 0.191x_1 + 0.015x_2 + 0.009x_3 + 0.739x_4 - 0.0014x_3x_4 + e.$$

Interactions between two categorical variables

It is the product between the dummy variables of both factors.

Example: In diamond example, consider x_1 the color and a new variable x_2 a factor for the certified origin (YES/NO). Then, (below without the weight)

$$\begin{aligned} y = & b_0 + b_E \delta_E(x_1) + b_F \delta_F(x_1) + b_G \delta_G(x_1) + b_H \delta_H(x_1) + b_I \delta_I(x_1) \\ & + a_N \delta_N(x_2) \\ & + c_{EN} \delta_E(x_1) \delta_N(x_2) + c_{FN} \delta_F(x_1) \delta_N(x_2) + c_{GN} \delta_G(x_1) \delta_N(x_2) \\ & + c_{HN} \delta_H(x_1) \delta_N(x_2) + c_{IN} \delta_I(x_1) \delta_N(x_2) + e \end{aligned}$$

Interactions between two categorical variables

```
> fit2 <- lm(Price ~ Weight+Color*Certif, data=diamond2)
> summary(fit2)
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2089.0	637.8	3.275	0.001182	**
Weight	11694.5	196.1	59.651	< 2e-16	***
ColorE	-3051.3	665.2	-4.587	6.65e-06	***
ColorF	-4137.3	639.6	-6.469	4.10e-10	***
ColorG	-4532.6	642.8	-7.052	1.26e-11	***
ColorH	-5103.2	635.0	-8.036	2.24e-14	***
ColorI	-5590.6	670.8	-8.334	3.00e-15	***
CertifYES	-2802.0	653.7	-4.286	2.46e-05	***
ColorE:CertifYES	1743.9	718.6	2.427	0.015831	*
ColorF:CertifYES	2672.6	687.4	3.888	0.000125	***
ColorG:CertifYES	2907.6	692.8	4.197	3.59e-05	***
ColorH:CertifYES	2964.0	689.2	4.300	2.32e-05	***
ColorI:CertifYES	2967.0	726.0	4.087	5.65e-05	***

```
---
[...]
```

Interactions between two categorical variables

- ▶ The number of model parameters is $1 + (g_1 - 1) + (g_2 - 1) + (g_1 - 1)(g_2 - 1)$, where g_j is the number of levels of factor j .
- ▶ This number is growing quickly with the number of levels.
- ▶ Interactions terms between categorical variables are often very difficult to interpret.

Higher order interactions

The interaction principle can be extended to higher order terms.
For example, two- and three way interactions,

```
> fit3 <- lm(Price ~ Weight*Color*Certif, data=diamond2)
```

```
> summary(fit3)
```

```
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	375.0	104154.5	0.004	0.997
Weight	13400.0	103635.0	0.129	0.897
ColorE	-5984.2	104161.6	-0.057	0.954
ColorF	-5058.4	104156.8	-0.049	0.961
ColorG	-4347.3	104158.0	-0.042	0.967
ColorH	-4493.6	104157.1	-0.043	0.966
ColorI	-4688.2	104164.4	-0.045	0.964
CertifYES	-3877.4	104155.4	-0.037	0.970
Weight:ColorE	3723.1	103644.4	0.036	0.971
Weight:ColorF	1757.0	103638.8	0.017	0.986
Weight:ColorG	235.3	103640.5	0.002	0.998
Weight:ColorH	-310.7	103639.0	-0.003	0.998
Weight:ColorI	-833.8	103646.2	-0.008	0.994
Weight:CertifYES	3057.0	103637.2	0.029	0.976
ColorE:CertifYES	7377.6	104163.0	0.071	0.944
ColorF:CertifYES	6354.3	104157.9	0.061	0.951
ColorG:CertifYES	6026.0	104159.2	0.058	0.954
ColorH:CertifYES	5971.8	104158.4	0.057	0.954
ColorI:CertifYES	6023.7	104165.8	0.058	0.954
Weight:ColorE:CertifYES	-8325.9	103647.8	-0.080	0.936
Weight:ColorF:CertifYES	-6466.1	103641.5	-0.062	0.950
Weight:ColorG:CertifYES	-6011.3	103643.3	-0.058	0.954
Weight:ColorH:CertifYES	-5816.6	103642.0	-0.056	0.955
Weight:ColorI:CertifYES	-5666.8	103649.3	-0.055	0.956

```
[...]
```

Interactions: interpretations and recommendations

Three and higher-way interactions should be used with care,

- ▶ They are very difficult to interpret
- ▶ The number of parameters in the model rapidly explodes with the number of variables (and with the number of levels).

Remember that the interaction between factor x_1 and x_2 is the modification of the effect of x_1 (on y) when x_2 is varying (and vice-versa).

In some contexts, interaction is also called moderation because " x_2 moderates the effect of x_1 ".

Interactions: using several models

If interaction between numerical and categorical variables means having several slopes, can't we use several separate regression?

- ▶ It is always possible to make several regressions (subgroup analysis)
- ▶ It is however not completely equivalent to one regression with a categorical variable and interaction.

The difference is that

- ▶ One overall model with interaction makes the assumption that the residuals e have the same structure: $e \sim N(0, \sigma^2)$, in all level of the categorical variable.
- ▶ With several models, each model can have a different structure of the residuals.

Interactions: using several models

When should we use separate regression and when one overall regression? Technical arguments:

- ▶ If standard deviation look inhomogeneous then several regressions is be justified.
- ▶ If standard deviation are homogeneous then a overall model (with the categorical factor!!) is better.

More later in the course about standard deviation analysis. In practice:

- ▶ If the difference between the grouping factor levels is of interest, then one has to use a single regression (with the categorical factor!!).
- ▶ If each subgroup is to be analyzed individually, then separate regression may be simpler.

Interactions: using several models

Examples:

- ▶ In a marketing study, behavior of women and men are both of interest in order to define a strategy of communication. Both are of interest and their difference is not of primary interest: the researchers use two separate regression models.
- ▶ In the study the equality in salary between women and men in a company is of interest. The difference between the two sexes is the central point. An overall model is needed.